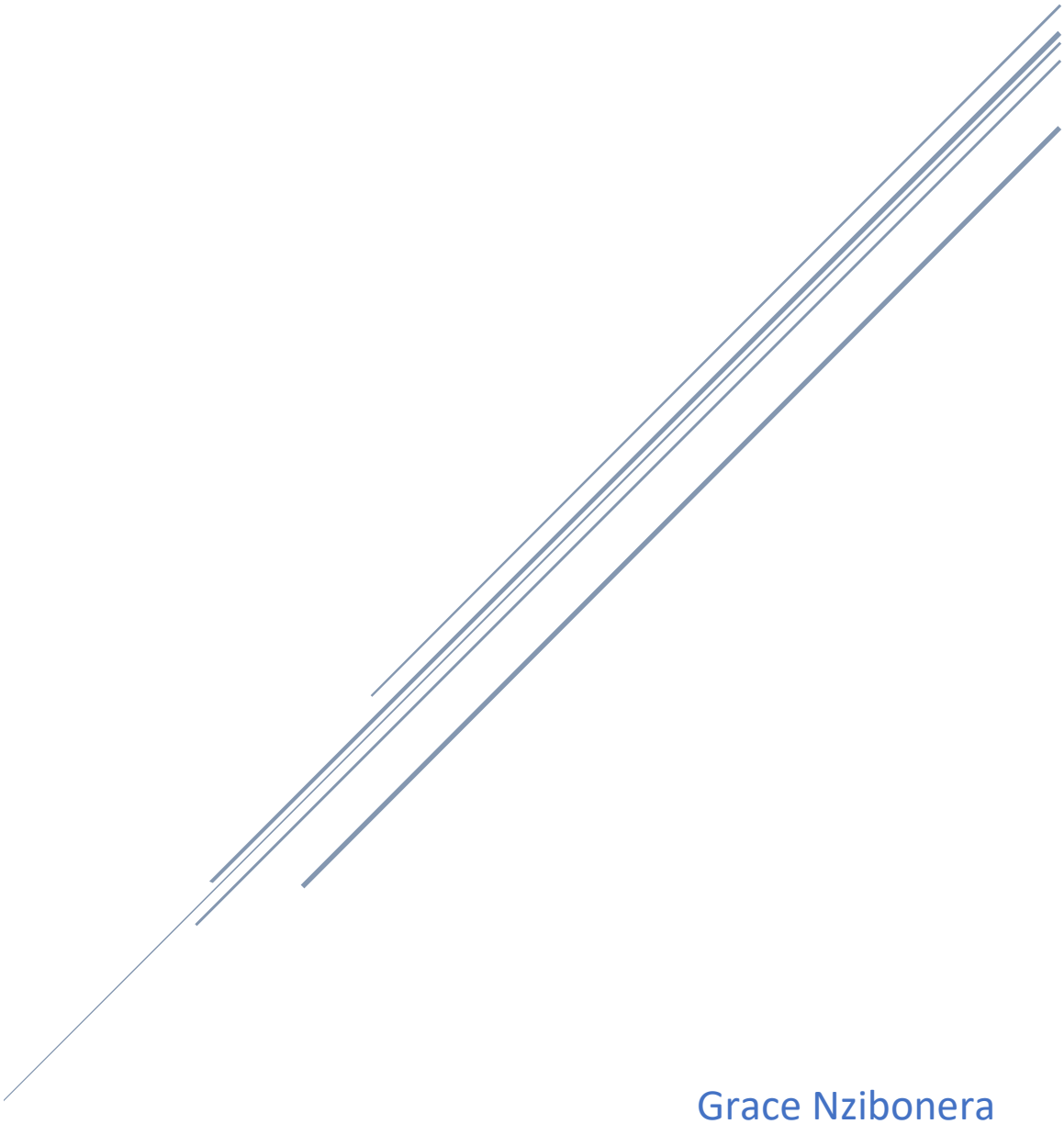
MULTIPLE LINEAR REGRESSION

Modelling Household Consumption Expenditure



Grace Nzibonera

DSCI 311

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1. Introduction

This report aims to evaluate the determinants of household consumption using a multiple linear regression model. It evaluates whether economic and financial factors such as household income, household wealth, prevailing interest and mortgage rates, consumer confidence index, local unemployment, years of education, local stock market index, and geographic region can explain variation in household consumption expenditure across 500 households. My analysis started with a full model as our baseline model, followed by a backward and forward stepwise selection to identify a set of predictors that have a significant explanatory power in the household consumption expenditure.

2. Model Selection

2.1 Full Regression Model

The multiple linear regression model explains household consumption very well. Overall, the model explains a substantial amount of the variation in consumption with a $R^2 = 0.809$, and an adjusted $R^2 = 0.805$ meaning about 81% of the variance in household consumption is accounted for by these predictors. Household income, household wealth, consumer confidence, mortgage rates, unemployment rates, and living in the South or West regions are significant predictors of consumption. Specifically, higher income, wealth, and consumer confidence index are associated with higher household consumption, while higher interest rates, mortgage rates, and unemployment rates are associated with lower consumption.

On the other hand, years of education, stock market returns, and living in the Northeast region do not meaningfully influence household consumption. Overall, the model is highly significant with a (**F-test $p < 0.001$**) which provides strong evidence that economic factors influence household consumption.

2.2 Backward Selection

Backward Selection is a model selection method that starts with all variables present in the model and and then removes the least significant one by one. Because it begins with the full model, it accounts for how variables behave together in the presence of all other predictors from the very start. In this analysis years of education and the local stock index turned out not to have any significant explanatory power in household consumption. This is evident because removing the two variables does not change the model performance implying that they were just noise. The model has **Adjusted R^2 of 0.805** compared to **0.8047** for the full model. Therefore, the reduced model is preferred given it has better complexity due to the fact that the model performance did not change despite having fewer predictors.

2.3 Forward Selection

Forward selection on the other hand starts with an empty model that is with the intercept only and adds the single most significant predictor one step-by-step. It evaluates variables based on how much explanatory power they add to what is already in the model. We start by adding the most significant variable to the least significant and stops when adding another variable does not improve the model performance. Forward selection arrived at the same seven predictors as backward selection, dropping years of education and local stock index. The conclusion remains the same that household wealth, consumer confidence, mortgage rates, unemployment rates, and living in the South or West regions are significant predictors of consumption with an overall fit of **$R^2 = 0.8085$**, and **adjusted $R^2 = 0.805$**Specifically, higher income, wealth, and consumer confidence index are associated with higher household consumption, while higher interest rates, mortgage rates, and unemployment rates are associated with lower consumption.

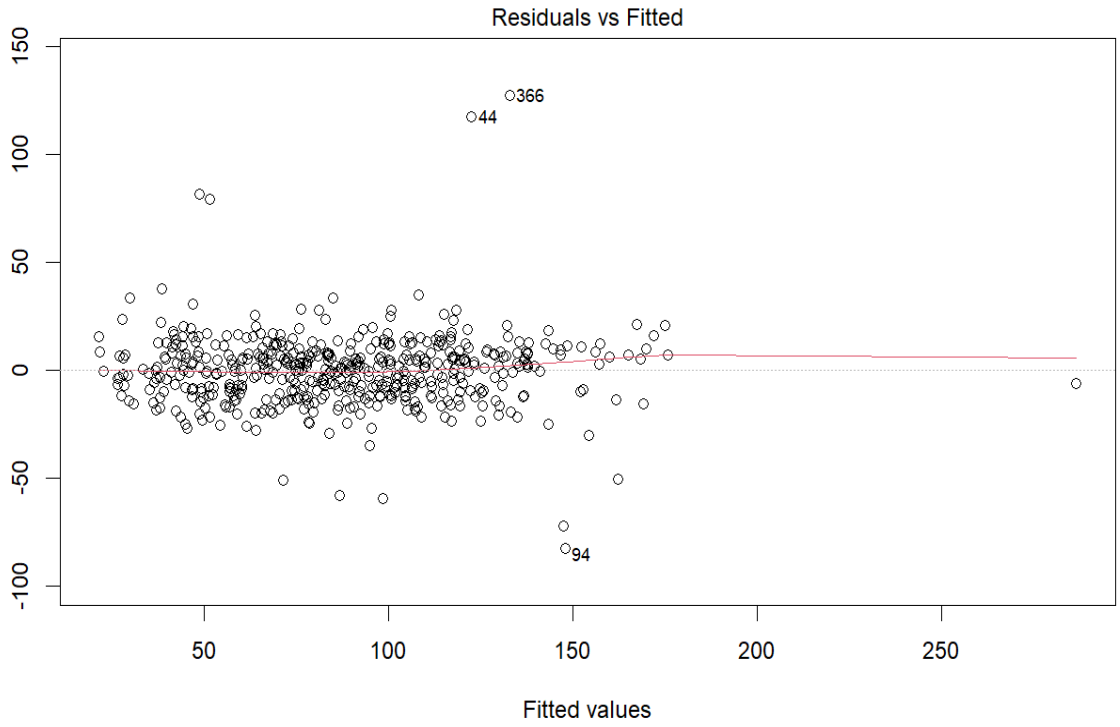
3. Model Comparison and selection

Metric	R-squared	Adj. R-squared	AIC	F-stat p-value	RSE
Full Model	0.809	0.804	4264	< 2.2e-16	16.97
Backward Selection	0.808	0.805	4261.4	< 2.2e-16	16.96
Forward Selection	0.808	0.805	4261.4	< 2.2e-16	16.96

The three candidate models were compared: the full model containing all 9 variables and the two forward and backward stepwise selection models which converged to identical models retaining 7 predictors. Household Income, Household Wealth, Interest Rate, Mortgage Rate, Consumer Confidence, Unemployment Rate, and Region — and dropping Years_of_Education and Local_Stock_Index_Return_Pct. The reduced models are preferred due to the fact that the model has better complexity, it performs nearly same as the full model but with a few predictors, which implies that years of education and local stock index are merely more noise than signal.

4. Model Diagnostics

4.1 Linearity

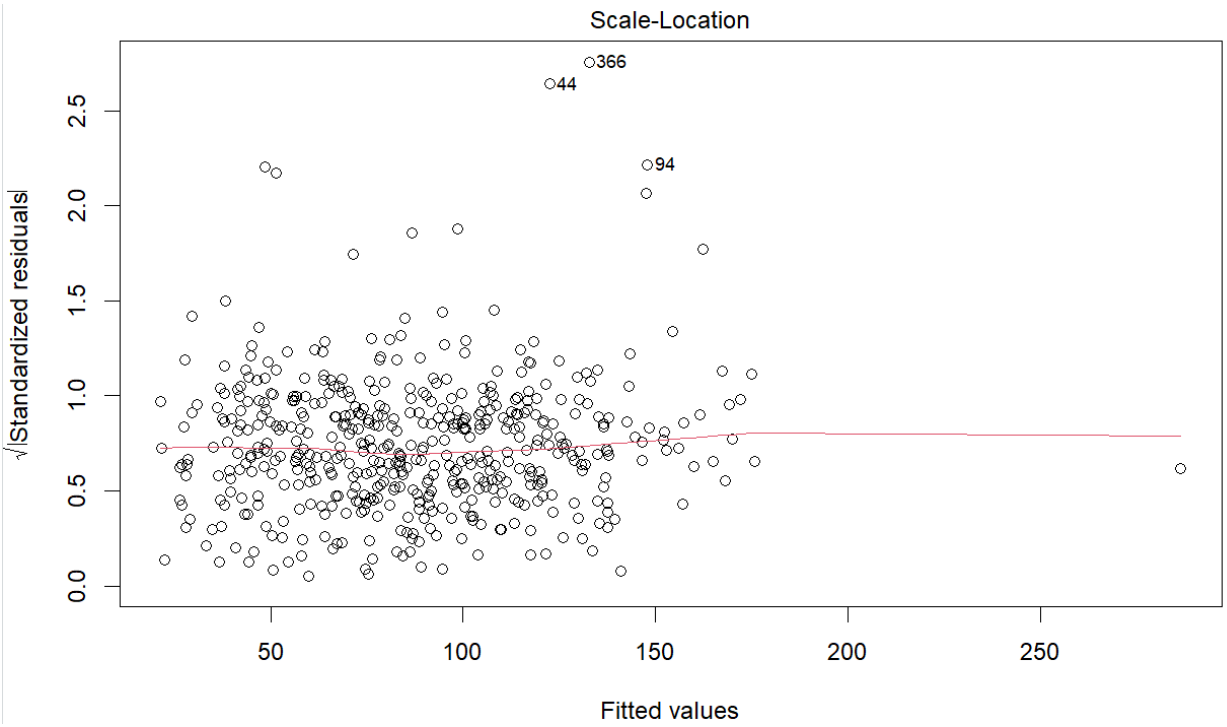


The graph above shows the linearity assumption for a regression model, which hold very well as shown above. The residuals are evenly scattered above and below zero with no systematic curvature, a few data points are flagged out above (point 366 and 44) which as we can see tend to pull the line up, they are worth noting but it's not a structural flaw.

4.2 Durbin-Watson test (Independence)

Durbin-Watson = 1.9993 with p = 0.4975 independence holds cleanly. The statistic sits almost exactly at 2.0, which is the value expected under no autocorrelation, and the p-value is nowhere near significant, so we fail to reject the null of no serial correlation. In plain terms: the residual for one household tells you nothing about the residual for the next, which is what the standard-error and p-value calculations assume.

4.3 Homoscedasticity

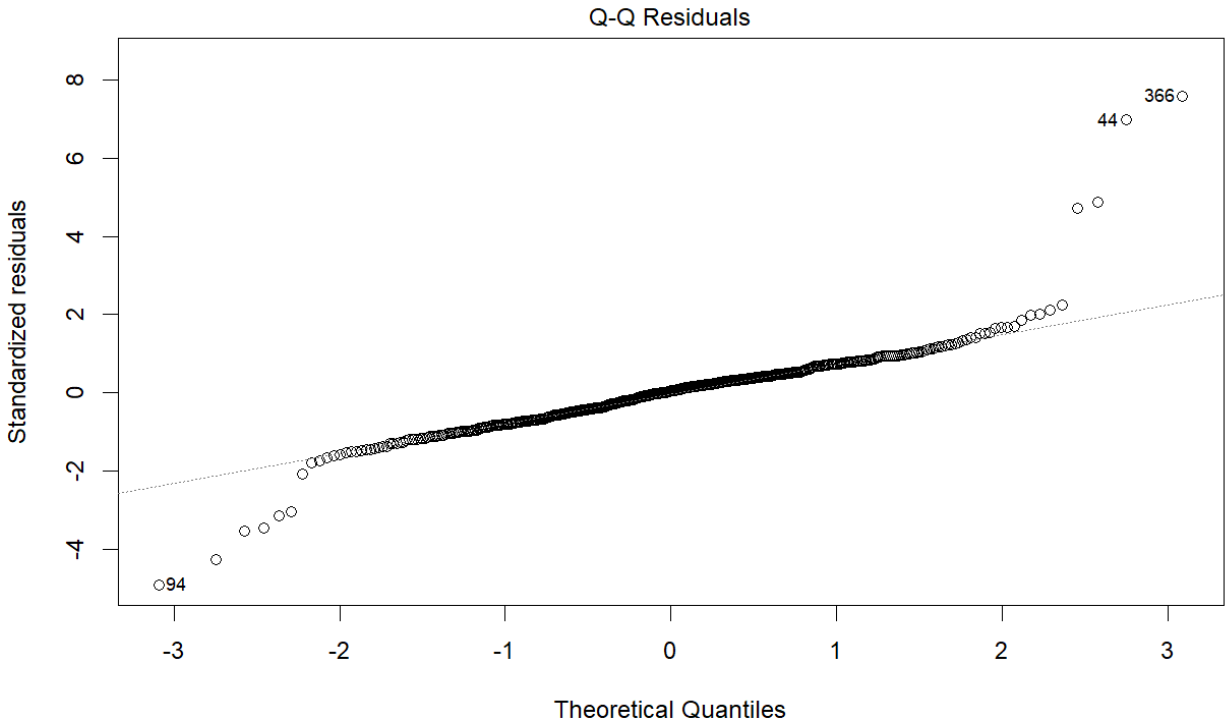


Homoscedasticity is another linear regression model assumption whereby we assume a constant variance to ensure that p-value, confidence intervals and hypothesis tests are valid. To examine this assumption, as we saw from linearity graph, the residuals were evenly scattered above and below zero with no funnel shape which suggests homoscedasticity. In addition, the scale-location plot backs that up, the red smoother is flat, not rising or falling, so dispersion stays roughly even across the fitted range. The three flagged points (366, 44, 94) sit high but they're individual outliers rather than a widening pattern, which is a different problem from heteroscedasticity.

4.3.1 Breusch-Pagan test

The studentized Breusch-Pagan test further confirms the above results, Breusch-Pagan = 14.256, df = 9, p = 0.1135. Homoscedasticity holds. The p-value is above 0.05, so we fail to reject the null of constant variance: there's no significant evidence to show that the spread of residuals changes with the fitted values. So, both the formal test (p = 0.1135) and the visual check agree: homoscedasticity holds.

4.4 Normality



The Q-Q plot shows normality fails, and it fails in a specific way worth naming precisely. The points track the reference line almost perfectly, which means the bulk of households are actually well-behaved. The deviation is entirely at the two ends: in the upper tail, points 366, 44, and a couple of others shoot far above the line and in the lower tail, point 94 falls well below the line. That pattern shows good fit in the centre, extreme points flying off at both ends is the signature of heavy tails, not skew alone. In practical terms, most households are predicted well, but a handful of extreme cases are predicted very badly, and those few are enough to reject normality for the model as a whole.

4.4.1 Shapiro wilk test

Shapiro-Wilk gives $W = 0.85097$, $p < 2.2e\text{-}16$ which means we reject the null hypothesis that the data are normally distributed, normality rejected. Shapiro is especially sensitive to exactly the pattern the Q-Q plot showed: a well-behaved centre with a few extreme values in the tails. Those three points (366, 44, 94) plus the handful of others at the ends are enough to push W down to 0.85, when a normally distributed set of residuals would give something close to 0.99.

5. Multicollinearity

Variables	VIF	Interpretation
Household Income K	3.48	Low multicollinearity
Household Wealth K	3.47	Low multicollinearity
Interest Rate Pct	2.93	Low multicollinearity
Mortgage Rate	2.93	Low multicollinearity
Consumer Confidence Index	1.03	Low multicollinearity
Unemployment Rate	1.01	Low multicollinearity
Region	1.02	Low multicollinearity

The VIF results indicate no serious multicollinearity concerns in the final model. The rule of thumb is VIF > 5 indicate potential concerns while VIF > 10 indicate serious multicollinearity problem.

All the VIF values are well below 5, with the highest being about 1.86, indicating that the predictors are not highly correlated with one another.

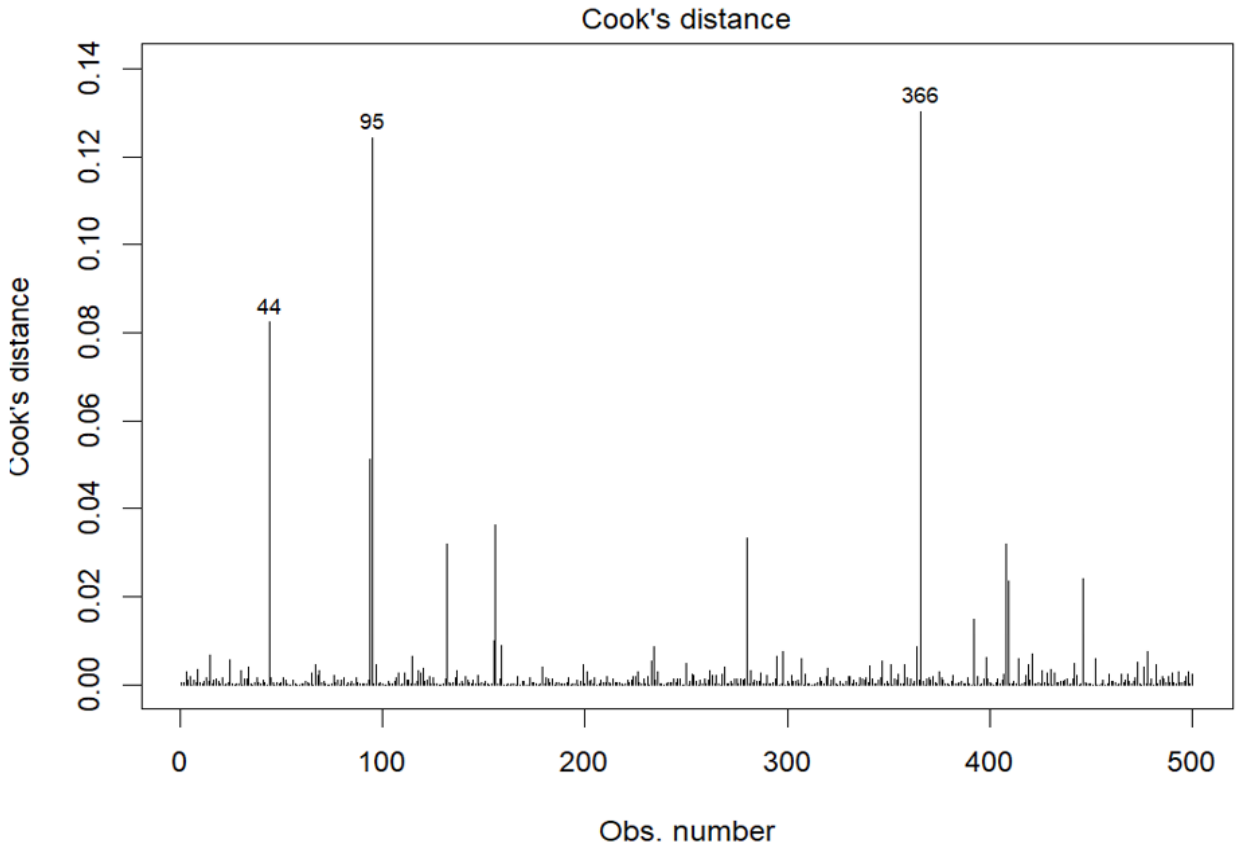
6. Conclusion

Assumption	Test	Result	Verdict
Linearity	Residuals vs Fitted	Flat smoother	✅
Independence	Durbin-Watson	DW = 1.9993, p = 0.4975	✅
Homoscedasticity	Breusch-Pagan	BP = 14.26, df = 9, p = 0.1135	✅
Normality	Shapiro-Wilk / Q-Q	W = 0.851, p < 2.2e-16	❌
Multicollinearity	VIF	max 3.48	✅

In conclusion, we choose the model that resulted from both forward and backward stepwise selection which arrived at the exact same model that household wealth, consumer confidence, mortgage rates, unemployment rates, and living in the South or West regions are significant predictors of consumption with an overall fit of **$R^2 = 0.8085$**, and **adjusted $R^2 = 0.805$.** Specifically, higher income, wealth, and consumer confidence index are associated with higher household consumption, while higher interest rates, mortgage rates, and unemployment rates are associated with lower consumption.

From the above table we can conclude that the coefficients themselves are still unbiased, and OLS remains the best linear unbiased estimator. What's affected is *inference*, the standard errors, t-statistics, and p-values assume normally distributed residuals, and heavy tails make them unreliable, particularly for the smaller coefficients like Interest Rate ($t \approx -2.75$) and Region: Northeast ($t \approx 1.53$). Those are the estimates where a normality violation could flip a conclusion. The natural next step would be examining the influential points which we evaluate by Cook's distance.

6.1 Cook's Distance



The rule of thumb for cook's distance, the threshhold is 4/n, which in our case is 4/500=0.008, a more serious concern is when Cook's Distance approaches **1**, most of the observations sit well below 0.008 with only a few data points (44, 95, and 366) exhibited larger Cook's Distance values relative to the rest of the sample. However, all values remained well below the commonly cited threshold of 1, suggesting that no observation exerted an unduly large influence on the regression estimates. Therefore, all observations were retained in the final model.

10. **Appendix**

10.1 Summary Statistics

summary(df)

```
 Household_ID       Household_Income_K Household_Wealth_K Interest_Rate_Pct Mortgage_Rate_Pct Consu
 Length:500         Min.   : 12.00     Min.   :   0.0     Min.   :0.250     Min.   : 1.000    Min.   : 40.00
 Class :character   1st Qu.: 58.99     1st Qu.: 148.7     1st Qu.:3.535     1st Qu.: 4.848    1st Qu.: 87.47
 Mode  :character   Median : 76.75     Median : 227.6     Median :4.660     Median : 6.115    Median :102.25
                    Mean   : 77.25     Mean   : 229.1     Mean   :4.555     Mean   : 6.042    Mean   :101.13
                    3rd Qu.: 95.28     3rd Qu.: 300.6     3rd Qu.:5.543     3rd Qu.: 7.125    3rd Qu.:115.06
                    Max.   :263.22     Max.   :1062.7     Max.   :9.000     Max.   :11.270    Max.   :172.20
 Unemployment_Rate_Pct Years_of_Education Local_Stock_Index_Return_Pct      Region    Household_Cons
 Min.   : 0.500        Min.   : 8.00      Min.   :-27.560                 Midwest  :122   Min.   : 15.50
 1st Qu.: 4.367        1st Qu.:12.29      1st Qu.: -1.060                 Northeast:111   1st Qu.: 59.88
 Median : 5.690        Median :13.71      Median :  6.350                 South    :142   Median : 85.08
 Mean   : 5.653        Mean   :13.88      Mean   :  6.469                 West     :125   Mean   : 87.36
 3rd Qu.: 7.080        3rd Qu.:15.46      3rd Qu.: 14.643                                 3rd Qu.:112.15
 Max.   :11.000        Max.   :22.00      Max.   : 40.530                                 Max.   :280.56
```

10.2 Full Model R Output

```
Residuals:
    Min      1Q  Median      3Q     Max
-82.718  -9.211   0.896   7.900 127.484

Coefficients:
                              Estimate Std. Error t value Pr(>|t|)
(Intercept)                  -9.19512    7.12169  -1.291  0.19726
Household_Income_K            0.50522    0.04666  10.827  < 2e-16 ***
Household_Wealth_K            0.12195    0.01148  10.618  < 2e-16 ***
Interest_Rate_Pct            -2.24662    0.80125  -2.804  0.00525 **
Mortgage_Rate_Pct            -4.94431    0.69123  -7.153 3.13e-12 ***
Consumer_Confidence_Index     0.71872    0.03524  20.394  < 2e-16 ***
Unemployment_Rate_Pct        -2.37936    0.38090  -6.247 9.14e-10 ***
Years_of_Education            0.09911    0.31309   0.317  0.75171
Local_Stock_Index_Return_Pct -0.07324    0.06708  -1.092  0.27542
RegionNortheast               3.44031    2.24603   1.532  0.12624
RegionSouth                  11.83932    2.10388   5.627 3.09e-08 ***
RegionWest                   21.71019    2.16706  10.018  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 16.97 on 488 degrees of freedom
Multiple R-squared:  0.8091,            Adjusted R-squared:  0.8047
F-statistic:   188 on 11 and 488 DF,  p-value: < 2.2e-16

cat("RMSE:", sqrt(mean(residuals(model_A)^2)), "\n")
RMSE: 16.76177
> cat("AIC:", AIC(model_A), "\n")
AIC: 4264.039
> cat("BIC:", BIC(model_A), "\n")
BIC: 4318.829
```

10.3 Backward Elimination R Output

```
Residuals:
    Min      1Q  Median      3Q     Max
-82.354  -9.335   0.802   7.970 127.386

Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)              -8.74991    5.48169  -1.596  0.11109
Household_Income_K        0.50375    0.04646  10.842  < 2e-16 ***
Household_Wealth_K        0.12261    0.01144  10.716  < 2e-16 ***
Interest_Rate_Pct        -2.20231    0.79979  -2.754  0.00611 **
Mortgage_Rate_Pct        -4.98808    0.68965  -7.233 1.83e-12 ***
Consumer_Confidence_Index  0.72309    0.03501  20.652  < 2e-16 ***
Unemployment_Rate_Pct    -2.37171    0.38059  -6.232 9.95e-10 ***
RegionNortheast           3.42500    2.24403   1.526  0.12759
RegionSouth              11.82772    2.10207   5.627 3.09e-08 ***
RegionWest               21.72407    2.16523  10.033  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 16.96 on 490 degrees of freedom
Multiple R-squared:  0.8085,  Adjusted R-squared:  0.805
F-statistic: 229.9 on 9 and 490 DF,  p-value: < 2.2e-16

> AIC(model_backward)
[1] 4261.408
> BIC(model_backward
```

10.4 Forward Selection R Output

```
Residuals:
    Min      1Q  Median      3Q     Max
-82.354  -9.335   0.802   7.970 127.386

Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)              -8.74991    5.48169  -1.596  0.11109
Household_Wealth_K        0.12261    0.01144  10.716  < 2e-16 ***
Consumer_Confidence_Index 0.72309    0.03501  20.652  < 2e-16 ***
Mortgage_Rate_Pct        -4.98808    0.68965  -7.233 1.83e-12 ***
RegionNortheast           3.42500    2.24403   1.526  0.12759
RegionSouth              11.82772    2.10207   5.627 3.09e-08 ***
RegionWest               21.72407    2.16523  10.033  < 2e-16 ***
Household_Income_K        0.50375    0.04646  10.842  < 2e-16 ***
Unemployment_Rate_Pct    -2.37171    0.38059  -6.232 9.95e-10 ***
Interest_Rate_Pct        -2.20231    0.79979  -2.754  0.00611 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 16.96 on 490 degrees of freedom
Multiple R-squared:  0.8085,  Adjusted R-squared:  0.805
F-statistic: 229.9 on 9 and 490 DF,  p-value: < 2.2e-16

> AIC(model_forward)
[1] 4261.408
> BIC (model_forward)
```

[1] 4307.769

10.5 R Code for the Entire Project

```
setwd("C:/Users/grace/OneDrive/Documents/School Stuff/Semester 4/Advanced Regression/Lab 1 assignment")

df <- read.csv("economics.csv")

View(df)

# STEP 1: Baseline Multiple Linear Regression

library(tidyverse)

# --- Load data ---

df <- read.csv("economics.csv", stringsAsFactors = FALSE)

df

# Set Region as a factor with Midwest as reference level

df$Region <- factor(df$Region, levels = c("Midwest", "Northeast", "South", "West"))

str(df)

summary(df)


# --- Fit the baseline model ---

model_A <- lm(Household_Consumption_K ~ Household_Income_K + Household_Wealth_K +
                Interest_Rate_Pct + Mortgage_Rate_Pct + Consumer_Confidence_Index +
                Unemployment_Rate_Pct + Years_of_Education + Local_Stock_Index_Return_Pct +
                Region,
              data = df)


summary(model_A)


# --- Key fit statistics ---

cat("R-squared:", summary(model_A)$r.squared, "\n")
```

```
cat("Adjusted R-squared:", summary(model_A)$adj.r.squared, "\n")
cat("RMSE:", sqrt(mean(residuals(model_A)^2)), "\n")
cat("AIC:", AIC(model_A), "\n")
cat("BIC:", BIC(model_A), "\n")

# --- Coefficient table with confidence intervals ---
confint(model_A)

# STEP 2a: Backward Selection


model_full <- lm(Household_Consumption_K ~ Household_Income_K + Household_Wealth_K +
                   Interest_Rate_Pct + Mortgage_Rate_Pct + Consumer_Confidence_Index +
                   Unemployment_Rate_Pct + Years_of_Education + Local_Stock_Index_Return_Pct +
                   Region,
                 data = df)
model_backward <- step(model_full,
                       direction = "backward",
                       trace = TRUE)

summary(model_backward)
AIC(model_backward)
BIC(model_backward)
# STEP 2b: Forward Selection


model_null <- lm(Household_Consumption_K ~ 1, data = df)

model_full <- lm(Household_Consumption_K ~ Household_Income_K + Household_Wealth_K +
```

```
            Interest_Rate_Pct + Mortgage_Rate_Pct + Consumer_Confidence_Index +
        Unemployment_Rate_Pct + Years_of_Education + Local_Stock_Index_Return_Pct +
            Region,
          data = df)

model_forward <- step(model_null,
                      scope = list(lower = model_null, upper = model_full),
                      direction = "forward",
                      trace = TRUE)

summary(model_forward)
AIC(model_forward)
BIC(model_forward)

# Model Diagnostic

library(lmtest)
library(car)
install.packages("tseries")
library(tseries)

model_selected <- lm(Household_Consumption_K ~ Household_Income_K + Household_Wealth_K +
                       Interest_Rate_Pct + Mortgage_Rate_Pct + Consumer_Confidence_Index +
                       Unemployment_Rate_Pct + Region,
                     data = df)

# 1. Linearity
plot(model_selected, which = 1)
```

```
# 2. Independence
dwtest(model_selected)

# 3. Homoscedasticity
bptest(model_selected)
plot(model_selected, which = 3)

# 4. Normality
plot(model_selected, which = 2)
jarque.bera.test(residuals(model_selected))
shapiro.test(residuals(model_selected))

# 5. Multicollinearity
vif(model_selected)

# 6. Outliers and influence
plot(model_selected, which = 4)
std_resid <- rstandard(model_selected)
df[abs(std_resid) > 3, ]
```